

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. CHB-401: Inorganic & Organic Chemistry-II

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Inorganic Chemistry-II – Marks: 35)

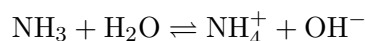
Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Why are alkali metals dissolved in liquid ammonia deep blue in color and good reducing agents?
- (b) Why are Ti^{3+} complexes colored, whereas Ti^{4+} complexes are colorless?
- (c) Why is the decrease in ionic size from Ce^{3+} to Lu^{3+} (lanthanide contraction) very small between adjacent elements?
- (d) Which one is a stronger Lewis acid: BCl_3 or $\text{B}(\text{CH}_3)_3$? Why?
- (e) Draw the structures of linkage isomers of $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]^{2+}$.

Question 2.

- (a) State the Brønsted-Lowry and Lewis concepts of acids and bases. Identify the conjugate acid-base pairs in:

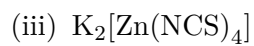


[6]

- (b) In terms of the HSAB principle, which end of the NCO^- (cyanate) ion would you expect to coordinate to Co^{2+} (borderline acid) and Ti^{4+} (hard acid)? Explain. [6]

Question 3.

- (a) Why do transition metals exhibit variable oxidation states? Which oxidation states are exhibited by Fe and Cr, and which ones are the most common? [6]
- (b) Write the IUPAC names of the following coordination compounds: [6]
 - (i) $[\text{Co}(\text{ONO})(\text{NH}_3)_5]^{2+}$
 - (ii) $[(\text{NH}_3)_5\text{Cr}-\text{OH}-\text{Cr}(\text{NH}_3)_5]\text{Cl}_5$



Question 4.

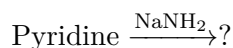
- (a) Compare and contrast the precipitation reactions in water and liquid ammonia solvents, giving two examples of each. [6]
- (b) Explain the important consequences of lanthanide contraction. Write the outer electronic configurations of Ho and Lu in the ground state. [6]

SECTION B

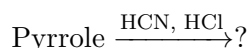
(Organic Chemistry-III – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) Why is pyrrole more reactive than benzene towards electrophilic substitution reactions?
- (b) Complete the following reaction:



- (c) Complete the reaction and write down the product:



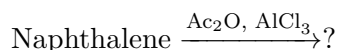
- (d) How will you convert D-glucose into D-fructose?
- (e) Define epimers and anomers of carbohydrates with examples.

Question 6.

- (a) Write the name of the following reaction and explain the formation of the products with a suitable mechanism: [6]



- (b) Complete the following reaction, showing the stepwise mechanism: [6]



Question 7.

- (a) Describe the Hantzsch pyridine synthesis with a detailed mechanism. [6]
- (b) Elucidate the structure of indigotin. How is it synthesized from anthranilic acid? [6]

Question 8.

- (a) How will you carry out the following conversions? [6]
- (i) Aldohexose to next higher aldohexose (Kiliani-Fischer synthesis)
- (ii) Aldohexose to next lower aldopentose (Wohl degradation)
- (b) Compare the electrophilic aromatic substitution reactions of furan, pyrrole, and thiophene. [6]

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